

WEBSITE TRAFFIC ANALYSIS

Field	Details
Project	Website Traffic Analysis
Tool	Python (Pandas, Matplotlib)
Dataset	Website Traffic Analysis — Kaggle
Analysis Period	19 August 2021 – 25 August 2021
Records	122,567 cleaned event records
Deliverables	Jupyter Notebook, analysis, visualizations, business insights & recommendations

Prepared for :- AlFido Tech Internship Program

Prepared by :- Ankit Kumar Mahto

Domain :- Data Analytics Intern

Dataset Source :- Kaggle — Website Traffic Analysis

1. Executive Summary

This report presents an analysis of 122,567 cleaned website traffic event records. The analysis uses Python, Pandas and Matplotlib to examine traffic activity, event types, geographic reach, content performance and link activity. The findings are converted into practical website optimization recommendations for AlFido Tech.

2. Dataset Overview

Metric	Value
Total records	122,567
Columns	9
Date range	19 Aug 2021 – 25 Aug 2021
Countries	211
Cities	11,993
Artists	2,419
Tracks	3,562
Unique links	3,839

3. Traffic KPI Summary

KPI	Value
Total events	122,567
Pageviews	73,360
Clicks	32,499
Previews	16,708
Pageview → Click event rate	44.30%
Pageview → Preview event rate	22.78%

4. Key Business Insights

- Pageviews are the largest recorded event category, making content discovery the dominant observable interaction.
- The dataset contains 32,499 click events and 16,708 preview events, showing substantial activity beyond pageviews.
- Traffic spans 211 countries and 11,993 cities, indicating broad geographic reach.
- Traffic volume alone should not be treated as a measure of engagement; traffic and interaction should be evaluated together.

- Popular artists, tracks and links provide opportunities for content prioritization, promotion and improved discovery.

5. Event-Level Traffic Funnel

The available events can be compared as an aggregate event funnel. This is not a user-level conversion funnel because the dataset does not contain user or session identifiers.

Stage	Events
Pageviews	73,360
Previews	16,708
Clicks	32,499

6. Five Website Optimization Recommendations

1. Improve Calls-to-Action (CTAs)

Make important actions clearer and easier to find. Use stronger CTA wording, better placement and simpler interaction paths. Track click events and event-level click rate.

2. Optimize High-Traffic, Low-Engagement Content

Compare traffic volume with click activity to identify content that attracts attention but needs stronger engagement. Prioritize optimization of these items.

3. Use Geographic Targeting

Prioritize markets using both traffic and engagement. Test localized content, regional campaigns and market-specific messaging.

4. Improve Content Discovery

Use related tracks, artists, trending content and regional recommendations to encourage deeper exploration of the content library.

5. Build a Continuous Traffic Dashboard

Create a Power BI or similar dashboard to monitor traffic, event mix, countries, artists, tracks and links on a recurring basis.

7. Recommended KPIs

- Total events
- Pageviews

- Clicks
- Previews
- Pageview-to-click event rate
- Pageview-to-preview event rate
- Traffic by country and city
- Artist and track activity
- Link performance

8. Data Limitations

The supplied dataset contains event-level records but does not contain user IDs, session IDs, event timestamps, page URLs or referral-source fields. Therefore, true unique users, sessions, bounce rate, average session duration, landing pages, exit pages and referral sources cannot be reliably calculated from this dataset. The project therefore uses event-level traffic and engagement proxies.

9. Conclusion

The website traffic analysis demonstrates broad international reach and substantial interaction across pageviews, previews and clicks. AlFido Tech can improve performance by focusing on engagement rather than traffic volume alone, optimizing calls-to-action, prioritizing valuable geographic markets, improving content discovery and continuously monitoring website KPIs.

10. Project Deliverables

- Cleaned dataset: traffic_cleaned.csv
- Jupyter Notebook: Task_3_Website_Traffic_Analysis.ipynb
- Editable Microsoft Word report: this document